1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Total trials	360 (train pool 324, hold-out 36)
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap
Sensor channels	144 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	5 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

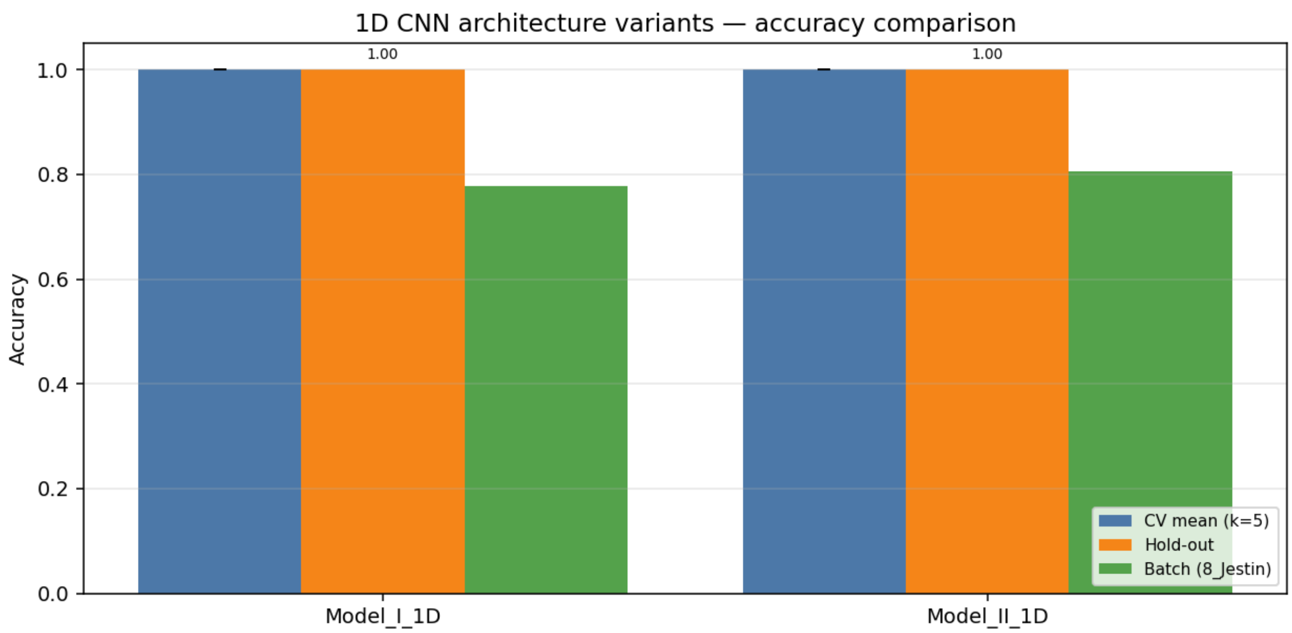
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

Enabled: 44 · Disabled: 0 · Modalities: YPR, Accel → 144 channels total.

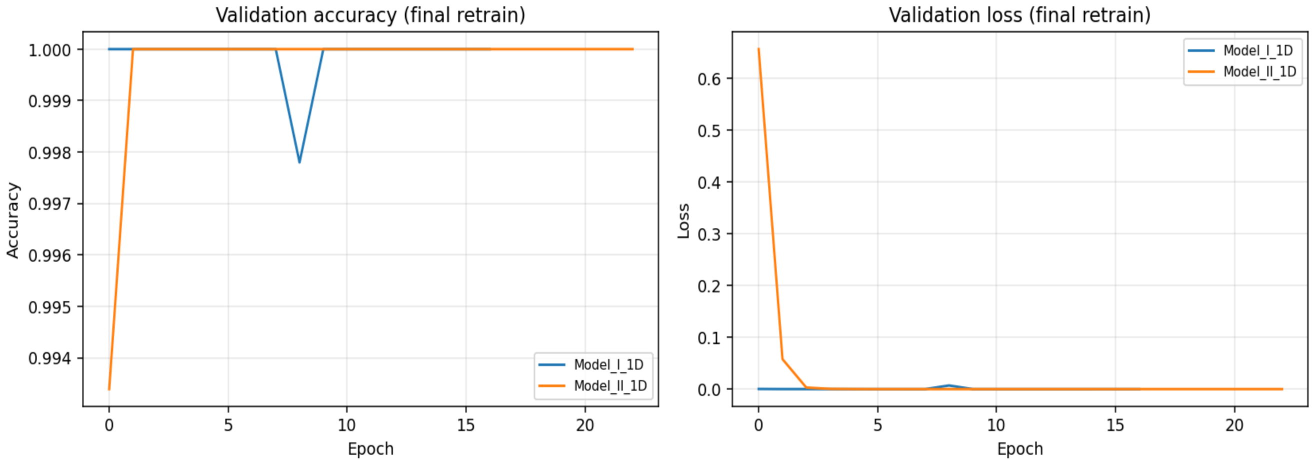
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	138,470	1.0000	0.0000	1.0000	0.0000	0.7778	28/36
Model_II_1D	93,766	1.0000	0.0000	1.0000	0.0000	0.8056	29/36

Best on hold-out: **Model_I_1D** · Best on CV: **Model_I_1D**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Model_I_1D
Model_II_1D
Validation loss (final retrain)
Loss
Epoch
Model_I_1D
Model_II_1D

Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	138,470
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7778 (28/36)

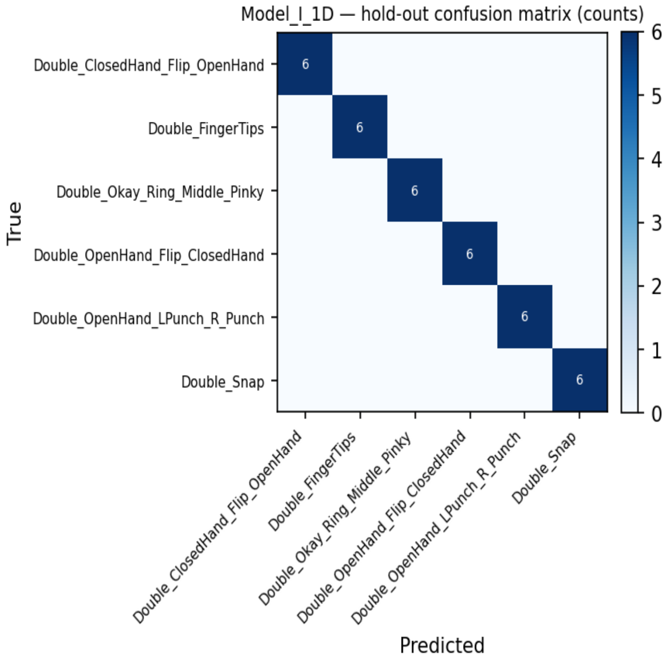
Cross-validation folds

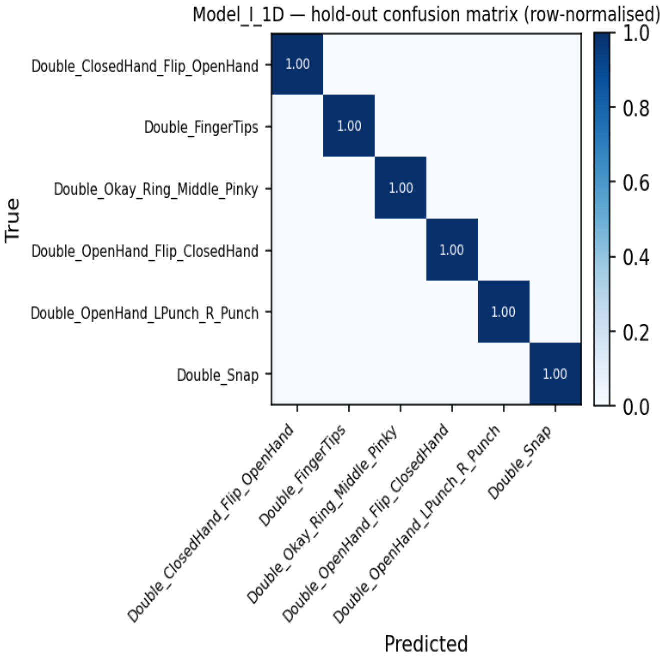
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1813	65	1.0000	1.0000	1.0000
2	1813	65	1.0000	1.0000	1.0000
3	1813	65	1.0000	1.0000	1.0000
4	1813	65	1.0000	1.0000	1.0000
5	1820	64	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

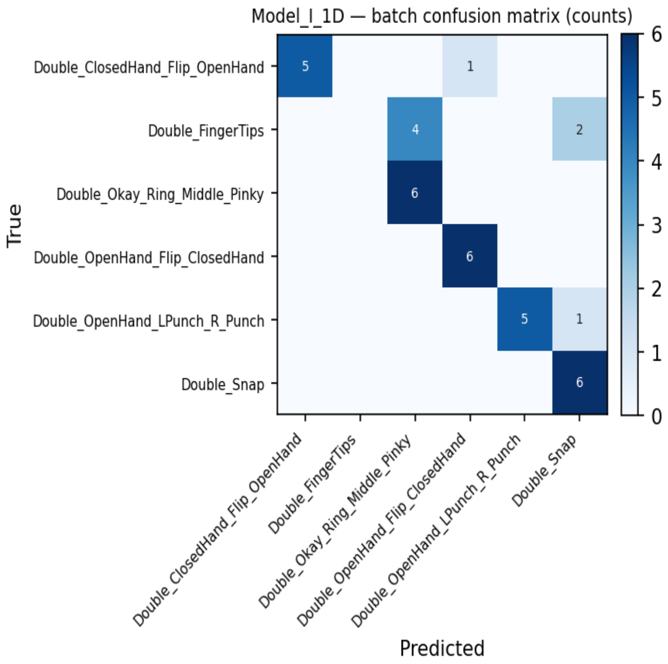




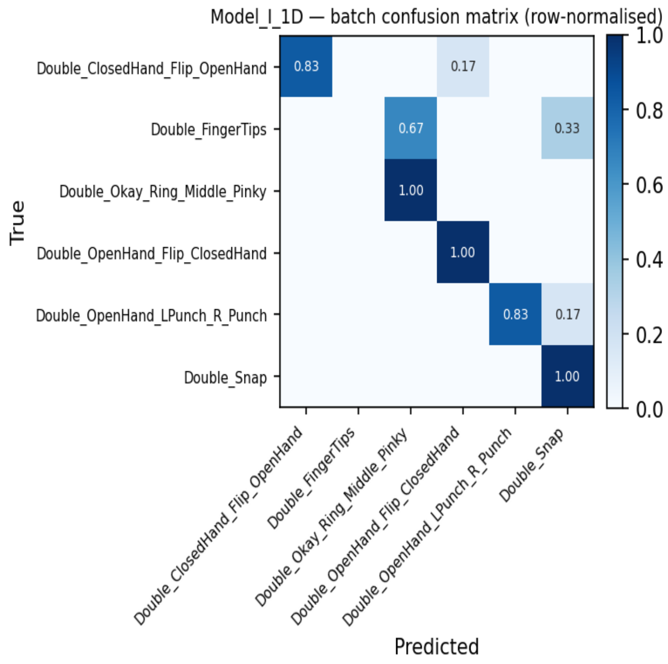
Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	0	6	0.000
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**28**	**36**	**0.778**

Batch confusion matrix (8_Jestin)



Model_I_1D — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
5
1
4
2
6
6
5
1
6
0
1
2
3
4
5
6



Model_I_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
0.83
0.17
0.67
0.33
1.00
1.00
0.83
0.17
1.00
0.0
0.2
0.4
0.6
0.8
1.0

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM):
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	93,766
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.8056 (29/36)

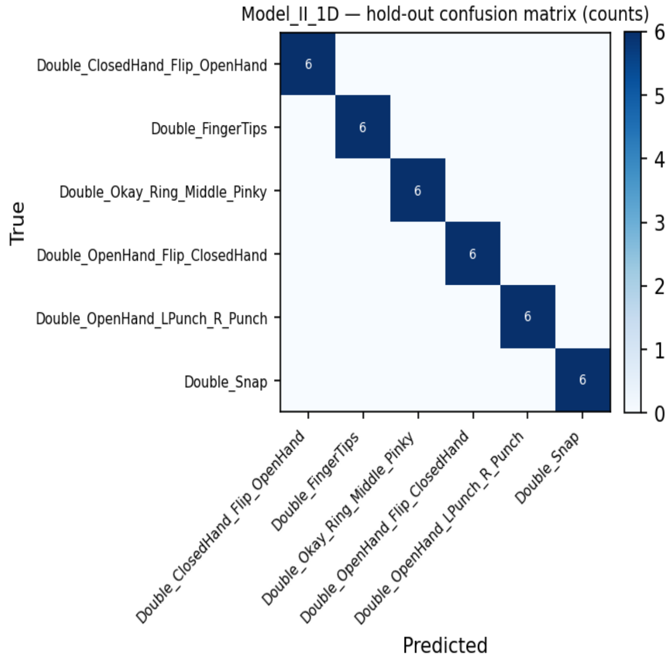
Cross-validation folds

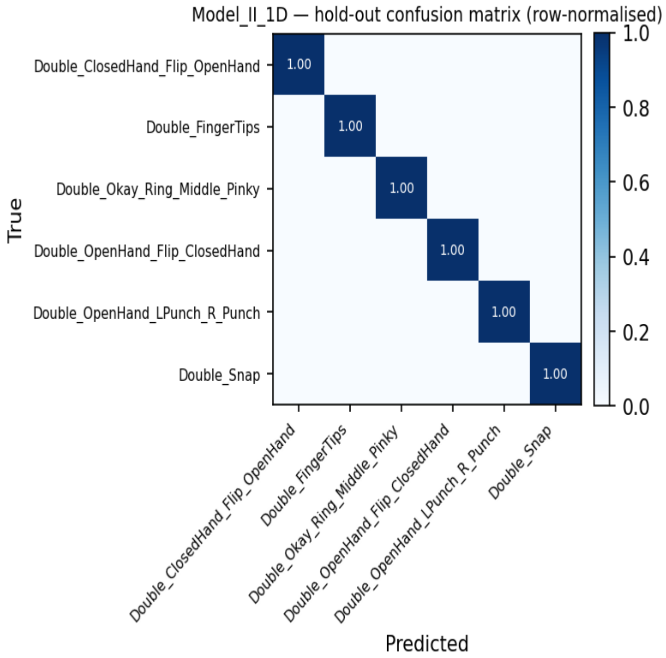
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1813	65	1.0000	1.0000	1.0000
2	1813	65	1.0000	1.0000	1.0000
3	1813	65	1.0000	1.0000	1.0000
4	1813	65	1.0000	1.0000	1.0000
5	1820	64	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

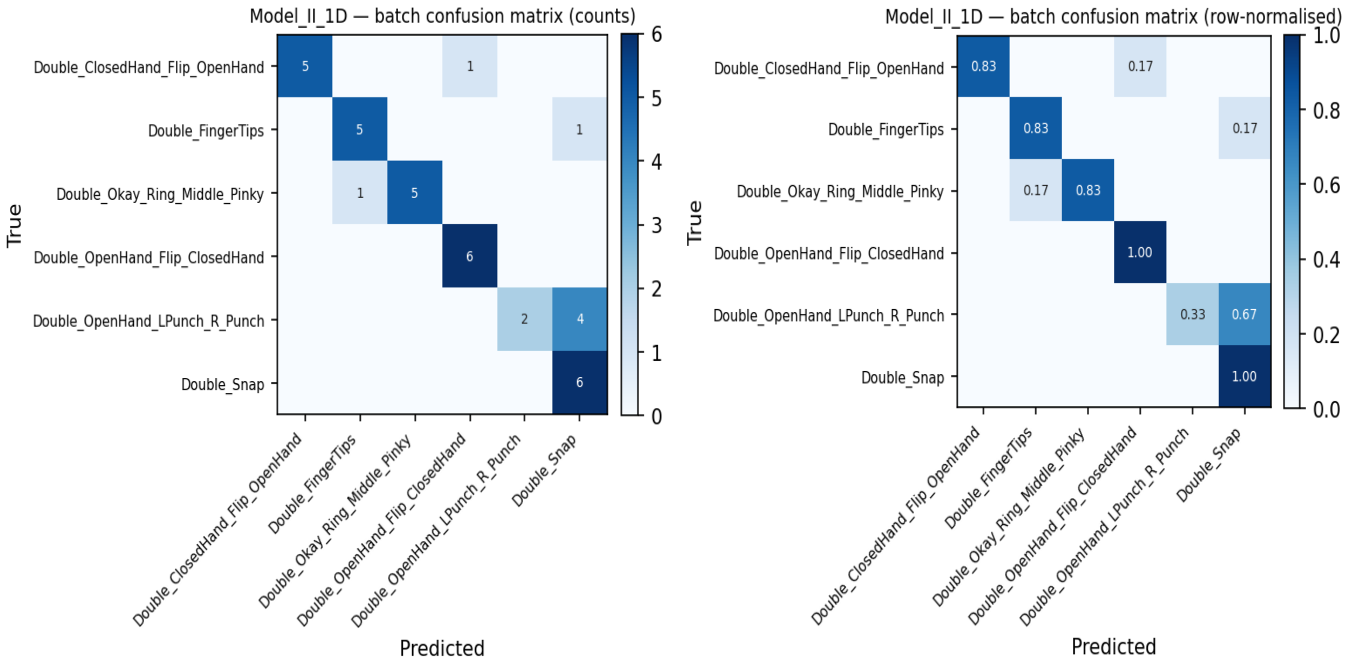




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	5	6	0.833
Double_Okay_Ring_Middle_Pinky	5	6	0.833
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	2	6	0.333
Double_Snap	6	6	1.000
Overall	**29**	**36**	**0.806**

Batch confusion matrix (8_Jestin)



Model_II_1D — batch confusion matrix (counts)
Model_II_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap